KPM-Bridge: An Uncertainty-Aware Cross-Implementation Telemetry Fabric for Portable AI xApps in Open 6G RAN

Yassir AL-Karawi and Hamed Al-Raweshidy

***Abstract*—Two Open Radio Access Network (O-RAN) implementations may emit valid E2 Service Model for Key Performance Measurement (E2SM-KPM) reports while assigning different meanings to the same artificial-intelligence (AI) xApp input. Units, entity scope, windows, clocks, counter resets, missingness, and filtering can all differ. KPM-Bridge treats this as a typed translation problem: it compiles source contracts, aligns event-time observations, learns an anchor-assisted map $M_v : y_{v,\le t} \mapsto \hat{z}_t$, and attaches a 48-byte evidence certificate. Split-conformal sets cover both $z_t \in \mathbb{R}^d$ and a registered xApp functional $\ell(z_t)$. On 201,912 public ColO-RAN samples with experiment-level separation, seven baselines, controlled profiles, and ablations, KPM-Bridge obtains 0.694 normalized root-mean-square error and 89.34% decision agreement across three stationary profiles; reductions relative to the strongest respective baselines are 2.6% and 0.53 percentage points. At nominal miscoverage $\alpha = .05$, joint coverage is 94.32%, while accepted actions have 0.209% disagreement at 54.45% acceptance. Under severe drift, the monitor detects 88.7% of shifted traces with 8.96 s median delay. Fail-closed invalidation changes post-shift disagreement exposure from 86.0% to 4.57% by reducing acceptance to 5.99%; conditional disagreement remains 76.23%. These profiles are controlled stressors, not commercial-vendor measurements.**

***Index Terms*—6G, Open RAN, E2SM-KPM, xApp portability, conformal prediction.**

I. Introduction

OPEN Radio Access Network (O-RAN) architectures disaggregate the radio access network (RAN), expose standardized interfaces, and host programmable control applications in a near-real-time RAN Intelligent Controller (Near-RT RIC) [1]–[3]. This arrangement is central to an artificial-intelligence (AI)-native 6G RAN: one xApp can observe several E2 nodes and adapt radio behavior without modifying every baseband stack [4]–[6]. The E2 Service Model for Key Performance Measurement (E2SM-KPM) standardizes the subscription and report structures transported over E2 [7].[1] Yet syntactic conformance, $\mathrm{decode}(y_{v,t}) = \mathrm{success}$, does not imply that reports from implementations $v \neq v'$ are interchangeable inputs to a learned model.

Suppose an xApp was trained on per-user-equipment downlink rate averaged over $w^\star = 250$ ms. A second stack reports a cumulative byte counter at cell scope every $w = 1$ s and resets it after restart. Both messages can decode and both may be labelled “throughput,” but a valid translation would still require differentiation, scope conversion, event-time alignment, and an account of lost information. The silent failure is precisely that the software path succeeds while the inferred action is evaluated outside the training semantics. Because KPM granularity is itself becoming an optimization variable in O-RAN digital-twin workflows [8], window w_t, age a_t, and support s_t cannot remain implicit metadata.

Open experimentation platforms have made AI xApps reproducible across controlled RAN environments [9]–[13]. xDevSM subsequently showed that application logic can be insulated from low-level service-model procedures and executed across heterogeneous open-source stacks [7], [14]. KPM-Bridge addresses the boundary after successful decoding. Protocol or application-programming-interface (API) portability asks whether an xApp can request and parse $y_{v,t}$; semantic portability asks whether $y_{v,t}$ denotes the expected quantity, entity, time support, and statistical regime. These are complementary conditions, not competing definitions.

The design rule is therefore simple: never expose a canonical vector $\hat{z}_t$ without its evidence state γ_t. The pair $(\hat{z}_t, \gamma_t)$ records the mapping identity, age, support, calibration epoch, uncertainty radius, and drift flags. A fixed xApp can then admit, defer, or abstain under explicit budgets $(a_{\max}, s_{\min}, \alpha)$ instead of treating every decoded report as equally reliable.

Five contributions follow from this rule.

- A machine-checkable contract types each stream by quantity, unit, entity, aggregation, window, clock, counter/reset semantics, schema version, and provenance. A minimum-cost assignment fails closed when no compatible source exists.
- A causal event-time design and anchor-assisted mapper represent lag, smoothing, loss, and resetting counters without relaxing the type constraints.
- Joint split-conformal sets $\mathcal{U}_t$ and functional intervals I_t^ℓ feed a selective gate and a calibrated multivariate residual monitor.
- Type soundness, translation-error decomposition, finite-sample coverage under stated exchangeability, decision stability, and online complexity are established formally.
- A deterministic trace-backed study compares seven portable baselines, a deployment retraining bound, five

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[1]Normative E2SM-KPM releases are distributed through the official O-RAN specification portal (accessed July 13, 2026); O-RAN specifications do not carry Crossref DOIs.

ablations, cluster-bootstrap intervals, and sensitivity to anchor rate, missingness, lag, miscoverage, and drift.

The contribution is Open-RAN-specific because the contract key is $(\mathrm{E2Node}, \mathrm{E2SMRev}, \mathrm{MeasID}, \mathrm{ModelID})$ across independently implemented producers. KPM-Bridge certifies compatibility and bounded decision uncertainty under stated assumptions. It does not prove that the xApp objective is safe, nor does it authenticate compromised code; execution integrity and interface security remain orthogonal [15].

II. Related Work

O-RAN service-model implementations expose KPM subscription, indication, and measurement information structures through E2 [3], [7]. General telemetry architectures separate acquisition, export, analysis, and consumption [16], [17]; the Network Management Datastore Architecture likewise emphasizes explicit operational state and provenance [18]. KPM-Bridge specializes these principles for learned Near-RT RIC inputs by treating scope, time support, counter semantics, and calibration epoch as components of the feature type rather than auxiliary documentation.

ColO-RAN demonstrated closed-loop machine-learning xApps on Colosseum and released traces spanning seven base stations, 42 user equipment (UE) devices, three slices, and three schedulers [9]. OpenRAN Gym added repeatable collection and testing workflows [10], [11]; FlexRIC, OrchestRAN, Open AI Cellular (OAIC), and ns-O-RAN broadened access to software development, orchestration, testbeds, and simulation [12], [13], [19], [20]. These platforms supply the experimental backbone used here, but they do not define a cross-implementation uncertainty contract for $(y_{v,t}, z_t)$.

The boundary becomes more important in native-AI digital twins, where E2/A1/O1 services support shadow tests, guarded control, and rollback [21]. Such a twin consumes telemetry and stresses portability, but it does not itself supply a typed KPM mapping, a finite-sample set for translation error, or a drift-aware abstention rule.

Open-RAN intelligence frameworks establish the roles of data-driven xApps and coordinated model placement [4], [5], [20]. FlexRIC offers a portable software-development kit (SDK) at the RIC boundary [19], while xDevSM hides service-model mechanics behind developer-facing APIs [7]. Its 2026 extension unifies observability and control across heterogeneous OpenAirInterface (OAI)- and srsRAN-based deployments and documents stack-specific granularity and scheduler behavior [14]. That work is the closest complement: KPM-Bridge can sit behind the API and test whether two decoded metrics are fit for the same registered input z_k, without replacing the API or disputing its protocol-portability claim.

Table I separates the condition "the API returns a value" from "the value is a valid realization of feature z_k." KPM-Bridge operates only on the latter boundary and therefore complements, rather than replaces, E2 decoders and portable xApp APIs.

Domain-adaptation bounds combine source error, distribution divergence, and a joint-labeling term that marginal alignment cannot remove [22]. Correlation alignment (CORAL), adversarial adaptation, and optimal transport (OT) reduce distribution mismatch [23]–[26], but they cannot decide whether two columns share a physical quantity and entity. An unconstrained map may reduce a divergence $D(P_s, P_t)$ while aligning cell load to per-UE rate. KPM-Bridge therefore permits statistical correction only after the type predicate is true.

Conformal prediction turns held-out nonconformity scores into finite-sample sets under exchangeability [27], [28]; weighted variants address quantified departures from exchangeability [29]. Reject-option and set-valued classifiers formalize the error–coverage tradeoff when abstention is allowed [30], [31]. Drift methods characterize and detect invalid operating regimes [32]–[36]. KPM-Bridge brings these elements to one O-RAN telemetry boundary, while keeping vector coverage $z_t \in \mathcal{U}_t$ distinct from functional coverage $\ell(z_t) \in I_t^\ell$.

III. Problem Formulation

Let $\mathcal{V}$ be the connected RAN implementations and $z_t = (z_{t,1}, \ldots, z_{t,d}) \in \mathcal{Z} \subseteq \mathbb{R}^d$ the canonical KPM state at event time t. The benchmark uses $d = 8$, but no result depends on that value. Multi-metric controllers may jointly consume queue, energy, latency, signal-to-interference-plus-noise ratio, and fairness observations, as in energy-aware off-grid O-RAN control [37]; compatibility must therefore be checked per feature and scope. Raw stream i from implementation $v \in \mathcal{V}$ has contract

$$c_{v,i} = (q_{v,i}, u_{v,i}, e_{v,i}, a_{v,i}, w_{v,i}, \tau_{v,i}, \rho_{v,i}, \nu_{v,i}), \tag{1}$$

where q denotes quantity, u physical unit, e entity scope, a aggregation, $w > 0$ observation window, τ clock basis, ρ gauge/delta/cumulative and reset semantics, and ν provenance plus schema version. Canonical coordinate $k \in \{1, \ldots, d\}$ has target $c_k^\star$.

Definition 1 (Type-compatible candidate)**.** *A source $c_{v,i}$ may supply $c_k^\star$ only if quantity, physical dimension, and entity scope agree. A unit difference creates an explicit conversion. Differences in aggregation, window, clock, or counter semantics create required transforms and residual uncertainty, not implicit equivalence.*

Let $m_{v,t} \in \{0,1\}^{p_v}$ be the observation mask for the p_v source streams. The report model is

$$y_{v,t} = m_{v,t} \odot \left[g_v(\Phi_v(z_{\le t})) + \epsilon_{v,t}\right], \tag{2}$$

where $\odot$ is elementwise multiplication, Φ_v applies windowing and delay, g_v contains declared unit/counter operations plus an identified residual map, and $\epsilon_{v,t}$ collects noise and quantization. Neither g_v^{-1} nor Φ_v^{-1} is assumed to exist: aggregation and missingness may destroy information.

KPM-Bridge returns

$$B_v(y_{v,\le t}) = (\hat{z}_t, \mathcal{U}_t, \gamma_t), \tag{3}$$

where $\mathcal{U}_t \subseteq \mathbb{R}^d$ is calibrated and γ_t contains schema hash, mapping identifier, calibration epoch, support, age, radius, and drift flags. The point map is $M_v(y_{v,\le t}) = \hat{z}_t$. For a fixed xApp functional $\ell : \mathbb{R}^d \to \mathbb{R}$ and threshold θ, an action is admitted only if its certified interval I_t^ℓ lies wholly in $(-\infty, \theta)$ or (θ, ∞).

The mapper is learned from paired anchors $\mathcal{D}_v^A = \{(y_{v,t}, z_t)\}$ and an independent calibration set $\mathcal{D}_v^C$, with

TABLE I
COMPARISON OF REPRESENTATIVE O-RAN TELEMETRY AND PORTABILITY WORK BY INTERFACE SCOPE, KPM SEMANTICS, TEMPORAL REPAIR, UNCERTAINTY SETS, AND DRIFT HANDLING.

Work	Primary layer	Cross-stack E2/API	Typed KPM meaning	Temporal/counter repair	Finite-sample set	Drift-aware abstention
O-RAN/E2SM studies [3], [7]	architecture/service model	standardized protocol	measurement catalogue	report semantics	no	no
ColO-RAN/OpenRAN Gym [9], [11]	experimentation	platform adapters	dataset-specific	pipeline-specific	no	no
ns-O-RAN [12]	RIC–simulation integration	yes	scenario-specific	simulator clock	no	no
FlexRIC/OrchestRAN [19], [20]	SDK/orchestration	yes	model-defined	runtime-specific	no	no
xDevSM [7], [14]	developer API	yes	unified metric access	service-model handling	no	no
KPM granularity xApp [8]	digital-twin reporting	xApp-level	granularity objective	adaptive granularity	no	no
KPM-Bridge	AI-input evidence layer	complementary	quantity/scope/unit contract	explicit event time and reset	joint and functional	yes

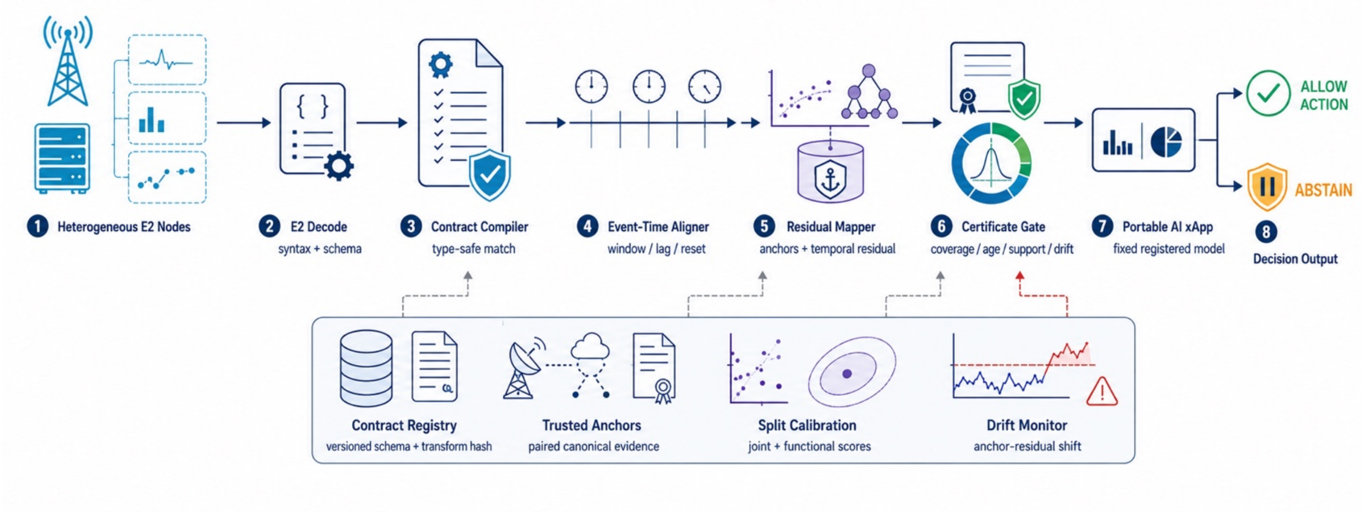

Fig. 1. KPM-Bridge placement between E2 decoding and a portable AI xApp. The data path compiles and aligns reports; the evidence plane supplies versioned contracts, trusted anchors, calibration, and drift state.

$\mathcal{D}_v^A \cap \mathcal{D}_v^C = \varnothing$. Let $D = \mathrm{diag}(s_1, \ldots, s_d) \succ 0$ scale the canonical coordinates. The fitting objective is

$$\min_{M_v} \sum_{(y,z)\in\mathcal{D}_v^A} \left\| D^{-1}(M_v(y) - z) \right\|_2^2 + \lambda\Omega(M_v), \tag{4}$$

subject to the compiled input plan. Here $\lambda \geq 0$ controls regularization and $\Omega(M_v)$ penalizes complexity. The benchmark sets s_k to the larger of the fit-prefix standard deviation and interquartile range divided by 1.349, with $s_k = 1$ only for a degenerate coordinate. Thus (D, λ) are determined without calibration or test data. Equation (4) begins only after semantic feasibility has been established. Operationally, the goal is not to maximize acceptance at any cost; it is to minimize canonical-action disagreement conditional on admission while reporting $\Pr\{\text{abstain}\}$.

The shift family includes declared unit and counter changes, window/lag, noise, quantization, entry loss, hidden monotone filtering, and post-deployment mean shift. We assume that E2 termination authenticates message origin. Malicious declarations, compromised xApp execution, and adversarial anchor corruption remain outside scope; provenance fields can bind external protections but cannot substitute for them.

IV. ARCHITECTURE AND ALGORITHMS

Figure 1 places the bridge after E2 decoding and before the portable xApp feature interface. The fast path $y_{v,t} \mapsto (\hat{z}_t, \gamma_t)$ stays inside the Near-RT RIC and adds no radio control round trip. A slower evidence plane maintains C_v, paired anchors, calibration scores, and the current validity state of epoch e.

A. Type-Safe Contract Compilation

Let $C_v = \{c_{v,i}\}_{i=1}^{p_v}$ and $C^\star = \{c_k^\star\}_{k=1}^{d}$ be the source and target contract sets. The compiler constructs a bipartite graph $G_v = (C_v, C^\star, E_v)$. An edge (i, k) exists only when quantity, physical dimension, and entity scope match; a missing edge is equivalent to cost $+\infty$. Every finite edge receives

$$\begin{aligned} \kappa_{ik} = \beta_a \mathbf{1}[a_i \neq a_k^\star] + \beta_w \left| \log \frac{w_i}{w_k^\star} \right|_{[0,K_w]} \\ + \beta_\tau \mathbf{1}[\tau_i \neq \tau_k^\star] + \beta_\rho \mathbf{1}[\rho_i \neq \rho_k^\star]. \end{aligned} \tag{5}$$

where $\boldsymbol{\beta} = (\beta_a, \beta_w, \beta_\tau, \beta_\rho) \in \mathbb{R}_+^4$ contains operator-set penalties. Since $w_i, w_k^\star > 0$, the log-ratio is defined, and $|x|_{[0,K_w]} = \min\{|x|, K_w\}$ with $K_w > 0$ prevents one window mismatch from dominating the matching objective. After type pruning, the compiler solves an injective matching $\pi_v : C^\star \hookrightarrow C_v$ minimizing $\sum_k \kappa_{\pi_v(k),k}$. Each selected edge emits unit/counter transforms, reset logic, clock/window obligations, and provenance binding. A window mismatch is not presumed invertible; it enlarges temporal and residual uncertainty. If $\pi_v(k)$ does not exist or $\kappa_{\pi_v(k),k}$ exceeds its budget, target k fails closed.

Figure 2 visualizes the two-stage rule in (5) and Algorithm 1. Quantity, dimension, and scope determine the Boolean type predicate $T_{ik} \in \{0, 1\}$; unit, counter, clock, and window differences affect κ_{ik} only after $T_{ik} = 1$. The red path is therefore a semantic rejection, not a low-confidence statistical prediction.

The schema hash and mapping identifier are disjoint slices of

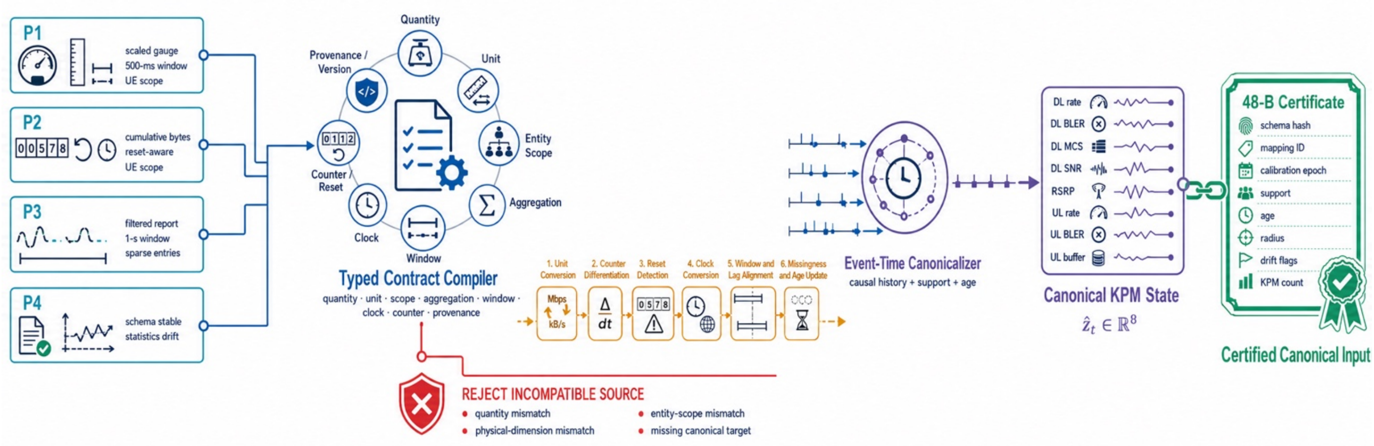

Fig. 2. Typed canonicalization from heterogeneous KPM profiles to a certified eight-dimensional state. Incompatible quantities or scopes fail closed; compatible streams receive unit/counter transforms and explicit clock, window, and missingness obligations.

Input: authenticated registry records C_v, targets $C^\star$, per-target budgets, version ν

1 Validate registry binding, mandatory fields, provenance, and schema versions.
2 For each (i, k), remove quantity, dimension, or scope mismatches; otherwise compute κ_{ik} by (5).
3 Solve the minimum-cost injective assignment over finite edges.
4 If any target is unmatched or exceeds its budget, return UNSUPPORTED with a reason code.
5 Emit target-ordered unit/counter transforms and clock/window obligations.
6 Compute one Secure Hash Algorithm 256 (SHA-256) digest over the canonical serialization of $(C_v, C^\star, \text{plan}, \nu)$ and its semantic budget; bind bytes 0–15 as the schema hash and bytes 16–23 as the mapping identifier.

Output: executable transform plan, or a fail-closed reason code

Algorithm 1: Compilation of source KPM contracts into a type-safe canonical transform plan.

one SHA-256 digest. This detail keeps the compiler, reference implementation, and fixed byte allocation in Table II consistent.

B. Event-Time Alignment and Residual Mapping

The executable plan applies declared transforms before any learned correction. For a cumulative counter b_t sampled after $\Delta_t > 0$, the candidate rate is $r_t = (b_t - b_{t-1})/\Delta_t$. If $b_t - b_{t-1} < 0$, the reset flag is raised. The alternative $r_t = b_t/\Delta_t$ is permitted only when reset metadata places the reset on the sampling boundary, as in the controlled profiles; otherwise $m_t^{(r)} = 0$. Missing adjacent values are not extrapolated.

For each canonical instant, the reference implementation forms a causal design vector

$$x_t = [\tilde{y}_t, \tilde{y}_{t-1}, \tilde{y}_{t-2}, \tilde{y}_{t-4}, \tilde{y}_{t-8}, \Delta\tilde{y}_t, m_t, \text{age}_t, \text{support}_t], \tag{6}$$

Here $\tilde{y}_t$ is the contract-normalized report, $\Delta\tilde{y}_t = \tilde{y}_t - \tilde{y}_{t-1}$ is causal, m_t is the mask, and $(\text{age}_t, \text{support}_t)$ summarize event-time staleness and completeness. The lag set is $\mathcal{L} = \{0, 1, 2, 4, 8\}$, so no future report enters x_t. One histogram gradient-boosted regressor per coordinate predicts the standardized target [38]. The released mapper uses 70 trees, at most 15 leaves, depth five, and fixed seeds. This estimator is replaceable: the contract and certificate interfaces also admit linear, monotone, or neural residual maps. A temporal-ridge variant appears in the ablation.

C. Joint and Functional Calibration

On held-out samples $(\hat{z}_j, z_j) \in \mathcal{D}_v^C$, define the joint nonconformity score

$$R_j = \left\| D^{-1}(\hat{z}_j - z_j) \right\|_2. \tag{7}$$

For n_c scores and $\alpha \in (0, 1)$, let

$$\begin{aligned} \hat{q}_\alpha &= R_{(\lceil (n_c+1)(1-\alpha) \rceil)}, \\ \mathcal{U}_t &= \{z : \left\| D^{-1}(z - \hat{z}_t) \right\|_2 \le \hat{q}_\alpha\}. \end{aligned} \tag{8}$$

The convention $R_{(n_c+1)} = +\infty$ preserves validity when $\alpha < 1/(n_c + 1)$. Under exchangeability, (8) is the standard split-conformal construction and gives $\Pr\{z_t \in \mathcal{U}_t\} \ge 1 - \alpha$ [27], [28]. A d-dimensional ball can nevertheless be conservative for one registered xApp, so KPM-Bridge separately calibrates

$$\begin{aligned} S_j &= |\ell(\hat{z}_j) - \ell(z_j)|, \\ \hat{q}_\alpha^\ell &= S_{(\lceil (n_c+1)(1-\alpha) \rceil)}, \\ I_t^\ell &= [\ell(\hat{z}_t) - \hat{q}_\alpha^\ell, \ell(\hat{z}_t) + \hat{q}_\alpha^\ell]. \end{aligned} \tag{9}$$

The functional interval supplements rather than replaces $\mathcal{U}_t$. It is bound to the registered model hash, so changing ℓ invalidates I_t^ℓ. The authenticated registry resolves $(\ell, \hat{q}_\alpha^\ell, \theta, a_{\max}, s_{\min})$ from the mapping identifier and calibration epoch; the compact sidecar need not repeat the fixed quantile. The guarantees $\Pr\{z_t \in \mathcal{U}_t\} \ge 1-\alpha$ and $\Pr\{\ell(z_t) \in I_t^\ell\} \ge 1-\alpha$ are marginal. Simultaneous coverage requires multiplicity correction or a joint score.

Drift invalidation and admission: Trusted anchors periodically reveal standardized residuals $r_t = D^{-1}(\hat{z}_t - z_t) \in \mathbb{R}^d$. Calibration anchors estimate (μ_r, Σ_r). For a window of $W = 5$ anchors, the monitor evaluates

$$H_t = \sqrt{W(\bar{r}_t - \mu_r)^\top (\Sigma_r + 0.1I)^{-1} (\bar{r}_t - \mu_r)}. \tag{10}$$

where $\bar{r}_t = W^{-1} \sum_{j=0}^{W-1} r_{t-j}$ and $0.1I_d$ regularizes the covariance. Anchors arrive every ten samples, or 2.5 s in this corpus. The alarm threshold $h_{.975}$ is the 97.5th percentile of the *per-stream maximum* calibration statistic, which accounts for repeated tests more directly than a pointwise quantile. Candidate W values are ranked without test data on held-out `exp1` streams under a predeclared 5% validation false-alarm ceiling and synthetic 1.25-scale shifts; this selects $W = 5$. If $H_t > h_{.975}$, the epoch is invalidated rather than the stale

Input: report/metadata, compiled plan, mapper, calibration epoch, quality budgets

1 Verify schema hash, mapping identifier, source provenance, and epoch.
2 Apply unit/counter transforms, enforce clock/window obligations, and update (6).
3 Predict $\hat{z}_t$ and attach $\hat{q}_\alpha$, support, age, and calibration epoch.
4 At an anchor, update H_t by (10); invalidate the epoch on a trigger.
5 Compute I_t^ℓ by (9) and evaluate support, age, drift, and threshold crossing.
6 Serialize the 48-byte certificate; emit the xApp action if admitted, else ABSTAIN with a reason code.

Output: canonical vector, certificate, action/abstention, and reason code

Algorithm 2: Certified streaming inference with event-time alignment, calibrated uncertainty, drift checks, and selective action.

interval widened. Recalibration is a separate control-plane operation.

Algorithm 2 is the online counterpart of Algorithm 1. Its admission predicate is the conjunction

$$G_t = \mathbf{1}\{\, \mathrm{age}_t \le a_{\max}, \mathrm{support}_t \ge s_{\min},\ H_t \le h_{.975}, \\ \theta \notin I_t^\ell,\ \gamma_t \text{ valid}\}. \tag{11}$$

Thus the temporal design in (6), functional interval in (9), and drift statistic in (10) meet in one auditable decision: act if $G_t = 1$, otherwise abstain with a reason code.

The encoded sidecar is fixed at 48 B: 16 bytes for the schema hash, 2×8 bytes for mapping and calibration identifiers, 3×4 bytes for support, age, and generic radius, and 2×2 bytes for flags and KPM count. The authenticated registry supplies the functional quantile, while the canonical vector is carried separately.

D. O-RAN Placement and Certificate Lifecycle

KPM-Bridge is an internal evidence layer, not a replacement for the E2 Application Protocol (E2AP) or E2SM-KPM. E2 termination handles subscription and Abstract Syntax Notation One decoding, then forwards values and metadata to the local adapter. The lookup key $(\mathrm{E2Node}, \mathrm{E2SMRev}, \mathrm{MeasID}, \mathrm{PolicyID})$ selects the contract. Service Management and Orchestration or the non-real-time RIC (Non-RT RIC) may provision the registry, but no new standardized A1, O1, or R1 message is claimed. The xApp-facing API returns $(\hat{z}_t, \gamma_t, \mathrm{reason}_t)$.

Figure 3 defines the persistent state machine $L_e \in \{\textsc{Unseen}, \textsc{Compiled}, \textsc{Calibrated}, \textsc{Active}, \textsc{Invalid}\}$. UNSEEN has no signed contract. COMPILED has a valid type match and transform plan but no statistical guarantee. CALIBRATED binds $(M_v, D, \hat{q}_\alpha, \hat{q}_\alpha^\ell, \mu_r, \Sigma_r)$ to epoch e. ACTIVE may evaluate G_t. INVALID is fail closed and retains its reason; it never transitions automatically to ACTIVE. A schema change returns to compilation, while a residual alarm requires new anchors and calibration.

Serialization uses network byte order and fixed-width fields. The 16-byte hash is a compact truncated digest, not a digital signature; registry updates must be authenticated by the RIC platform. Mapping and epoch identifiers prevent the invalid combination $(\hat{z}_t^{(e)}, I_t^{\ell,(e')})$ with $e \neq e'$. Floating-point support, age, and radius are adequate for the reference implementation, while a safety-critical deployment may specify fixed point under a new layout version.

TABLE II
FIXED 48-BYTE CERTIFICATE LAYOUT AND THE OPERATIONAL MEANING OF EACH FIELD.

Field	Bytes	Purpose
Schema hash	16	binds source, target, and plan
Mapping identifier	8	selects fitted mapper
Calibration epoch	8	invalidation boundary
Support	4	observed KPM fraction
Age	4	event-time staleness in ms
Radius	4	generic conformal radius
Flags	2	drift/schema/quality bits
KPM count	2	canonical dimension guard
Total	48	network byte order

Table II accounts for all $16+8+8+4+4+4+2+2 = 48$ bytes. The object is a local evidence sidecar; $\hat{z}_t$ remains a separate payload.

Reason codes separate $\mathrm{unsupported}$ from $\mathrm{uncertain}$. More samples cannot repair $T_{ik} = 0$, whereas a type-compatible stream may recover after a fresh report or anchor. Table III maps each predicate failure to a fail-closed action and a distinct recovery path, preventing an orchestrator from repeatedly refining a fundamentally incompatible KPM.

V. ANALYTICAL GUARANTEES

Proposition 1 (Compiler type soundness)**.** *If Algorithm 1 returns a mapping, every selected source and target have equal quantity, physical dimension, and entity scope.*

Proof: An edge is inserted only after all three equality predicates pass. The assignment solver selects only finite inserted edges. Therefore every returned pair satisfies the predicates. Unit, temporal, and counter differences may remain, but they are explicit transforms rather than silent type substitutions. ■

Type soundness is necessary but not sufficient for identifiability. A cell mean cannot generally recover a per-UE vector, even when both are expressed in Mbit/s, so $T_{ik} = 0$ rejects that scope change. By contrast, one-second and 250-ms means may satisfy $T_{ik} = 1$ while carrying unequal information. Their edge is admissible only with an explicit window transform and residual uncertainty.

Proposition 2 (Translation-error decomposition)**.** *Unless a subscript states otherwise, the norms in this proposition are Euclidean. Let u_t^0 be the noise-free, full-support output of the declared transform; u_t^m its masked version; and u_t^ϵ the noisy masked input. Let F_v^0 be the fitted calibration-regime realization of M_v, and let $\bar{z}_t^0$ and $\bar{z}_t$ be the information-preserving targets under the calibration and current regimes. Then $\hat{z}_t = F_v^0(u_t^\epsilon)$ obeys*

$$\|\hat{z}_t - z_t\| \le E_{\mathrm{temp}}(t) + E_{\mathrm{model}}(t) + E_{\mathrm{noise}}(t) \\ + E_{\mathrm{miss}}(t) + E_{\mathrm{drift}}(t), \tag{12}$$

where $E_{\mathrm{noise}} = \left\|F_v^0(u_t^\epsilon) - F_v^0(u_t^m)\right\|$, $E_{\mathrm{miss}} = \left\|F_v^0(u_t^m) - F_v^0(u_t^0)\right\|$, $E_{\mathrm{model}} = \left\|F_v^0(u_t^0) - \bar{z}_t^0\right\|$, $E_{\mathrm{drift}} = \left\|\bar{z}_t^0 - \bar{z}_t\right\|$, *and* $E_{\mathrm{temp}} = \|\bar{z}_t - z_t\|$.

Proof: Insert $F_v^0(u_t^m)$, $F_v^0(u_t^0)$, $\bar{z}_t^0$, and $\bar{z}_t$ between $\hat{z}_t$ and z_t, then apply the triangle inequality. Compiler soundness

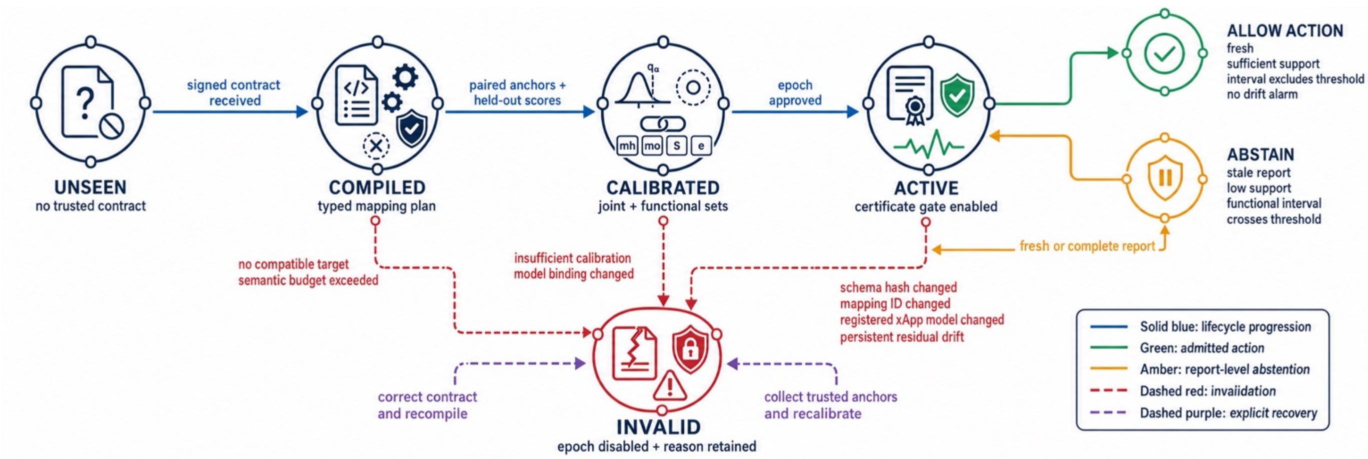

Fig. 3. Certificate lifecycle from an unseen contract to an active or invalid state. Schema or model changes force recompilation, whereas statistical drift requires new trusted anchors and recalibration.

TABLE III
FAIL-CLOSED CONDITIONS, EMITTED REASON CODES, AND REQUIRED RECOVERY ACTIONS.

Condition	Observable evidence	Immediate action	Recovery
Quantity/dimension/scope mismatch	no finite compiler edge	reject mapping	correct contract or source KPM
Unknown reset or adjacent counter loss	invalid delta and reduced support	mark feature missing	next valid delta or anchor
Schema/model hash change	certificate binding mismatch	invalidate epoch	recompile and recalibrate
Age or support budget violation	$\mathrm{age}_t > a_{\max}$ or $\mathrm{support}_t < s_{\min}$	abstain with reason	fresh/complete report
Functional interval crosses threshold	$\theta \in I_t^\ell$	abstain	refine, wait, or operator policy
Anchor-residual mean shift	H_t exceeds stream threshold	invalidate epoch	diagnose and recalibrate

removes a dimension/scope substitution term but not temporal or information-loss terms. ∎

Proposition 3 (Non-identifiability under aggregation)**.** *Suppose a source reports only $y = Az$, where A is a linear aggregation operator, and there exist $z^{(1)} \neq z^{(2)}$ in the admissible state space with $Az^{(1)} = Az^{(2)}$. For any deterministic point mapper $M(y)$, at least one of the two states has*

$$\left\| M(y) - z^{(j)} \right\|_2 \geq \frac{1}{2} \left\| z^{(1)} - z^{(2)} \right\|_2 . \tag{13}$$

Proof: Both states generate the same report and therefore the same estimate. If both errors were smaller than half their separation, the triangle inequality would imply $\|z^{(1)} - z^{(2)}\|_2 < \|z^{(1)} - z^{(2)}\|_2$, a contradiction. ∎

Proposition 3 quantifies the information that no mapper can recover: if $z^{(1)} - z^{(2)} \in \ker(A)$, both states produce the same y, and at least one point error is half their separation. Anchors or structural priors may choose among such states, but that dependence must be bound to the mapping identifier and uncertainty set. The scope predicate rejects the clearest version of this failure before fitting begins.

Assumption 1 (Calibration exchangeability)**.** *Conditional on a fixed schema, mapping, model, and non-drift regime, the test score and the n_c calibration scores are exchangeable. Calibration data are not used to fit the mapper.*

Proposition 4 (Finite-sample coverage)**.** *Under calibration exchangeability, the set in (8) satisfies*

$$\Pr\{z_{n_c+1} \in \mathcal{U}_{n_c+1}\} \geq 1 - \alpha. \tag{14}$$

The functional interval in (9) obeys the analogous inequality for $\ell(z)$.

Proof: The rank of the test nonconformity score among $n_c + 1$ exchangeable scores is uniform up to ties. The order statistic in (8) excludes at most an α fraction of ranks. Applying the same argument to S_j gives functional coverage [27], [28]. ∎

Radio traces are temporally dependent. The experiment separates repetitions and subsamples calibration points, but Proposition 4 remains conditional on exchangeability of $(R_1, \ldots, R_{n_c+1})$. Section VII therefore reports measured coverage under the implemented split rather than extending (14) to arbitrary autocorrelation.

Proposition 5 (Decision stability)**.** *For a threshold action $a(z) = \mathbf{1}[\ell(z) \geq \theta]$, if $\theta \notin I_t^\ell$ and $\ell(z_t) \in I_t^\ell$, then $a(\hat{z}_t) = a(z_t)$. Under calibration exchangeability, the joint probability of passing the interval gate and disagreeing with the canonical action is at most α; no conditional selective-error bound is claimed.*

Proof: When I_t^ℓ lies strictly above or below θ, all contained values induce the same action. Functional coverage gives $\Pr\{\theta \notin I_t^\ell,\, a(\hat{z}_t) \neq a(z_t)\} \leq \alpha$. Support, age, and drift filters preserve this joint bound because they only remove events. They do not imply the conditional bound $\Pr\{a(\hat{z}_t) \neq a(z_t) \mid \text{admit}\} \leq \alpha$; selective error is therefore measured separately. ∎

For a general utility $J : \mathcal{A} \times \mathcal{Z} \to \mathbb{R}$, a set-valued xApp can instead choose

$$a_t^\star \in \arg\max_{a \in \mathcal{A}} \min_{z \in \mathcal{U}_t} J(a, z), \tag{15}$$

following the robust-optimization principle [39]. The evaluated binary risk xApp uses I_t^ℓ, because the interval is tailored to

TABLE IV
CONTROLLED IMPLEMENTATION PROFILES APPLIED TO PUBLIC COLO-RAN TRACES; PROFILE NAMES DENOTE STRESS SETTINGS RATHER THAN VENDORS.

Profile	Declared representation	Window	Lag	Entry loss	Noise	Quantization	Additional condition
P1	scaled units / gauges	2 samples	1 sample	3%	2%	1%	mild hidden filter
P2	rates as resetting counters	3 samples	1 sample	8%	3%	1.5%	resets every 257 samples
P3	scaled units / gauges	4 samples	2 samples	18%	5%	3%	sparse filtered reports
P4	scaled units / gauges	2 samples	1 sample	5%	2.5%	1.2%	shift at 55% of each test trace

its registered threshold $a(z) = \mathbf{1}\{\ell(z) \geq \theta\}$.

For complexity, let $n_{\text{asn}} = p_v + d$ be the padded assignment size, $h = \dim(x_t)$, N_{tr} the trees per coordinate, L_{tr} their maximum depth, $N_{\text{leaf}} \leq 2^{L_{\text{tr}}}$ their maximum leaves, and W the drift window. Dense matching costs $O(n_{\text{asn}}^3)$ time and $O(p_v d)$ memory, but runs only after a schema change. Per report, declared transforms and feature updates cost $O(hd)$, tree inference $O(dN_{\text{tr}}L_{\text{tr}})$, and set construction $O(d)$. At an anchor epoch, H_t costs $O(d^2 + Wd)$. Persistent model storage is $O(dN_{\text{tr}}N_{\text{leaf}})$. Once $(\hat{q}_\alpha, \hat{q}_\alpha^\ell, \mu_r, \Sigma_r)$ are materialized, online cost is independent of n_c.

VI. Experimental Design

A. Trace Backbone and Controlled Profiles

The benchmark pins public ColO-RAN commit `bd86629` [9]. Its Rome-static-medium scenario contains seven base stations (BSs), 42 UEs, 10 MHz bandwidth, 50 physical resource blocks (PRBs), and enhanced mobile broadband (eMBB), machine-type communication (MTC), and ultra-reliable low-latency communication (URLLC) slices. Round-robin, water-filling, and proportional-fair schedulers span 28 resource allocations.

The deterministic subset is the Cartesian cross of three schedulers, configurations $\{\texttt{tr0}, \texttt{tr9}, \texttt{tr18}\}$, experiments $\{\texttt{exp1}, \texttt{exp2}\}$, BS1/BS4, and one UE per traffic class. It contains 108 files and 208,585 raw rows. A predeclared rule excludes two traces with fewer than 400 attached samples. After removing the four-sample prediction horizon, 53 `exp1` traces (109,539 rows) form the fitting/calibration source and 53 independent `exp2` traces (92,373 rows) form the test set. Every file is hash verified; GNU General Public License version 3.0 (GPL-3.0) data are fetched from upstream rather than redistributed.

The canonical vector is $z_t = (R_t^{\text{DL}}, B_t^{\text{DL}}, M_t^{\text{DL}}, S_t^{\text{DL}}, P_t^{\text{RSRP}}, R_t^{\text{UL}}, B_t^{\text{UL}}, Q_t^{\text{UL}}) \in \mathbb{R}^8$: reported downlink (DL) rate, DL block error rate (BLER), DL modulation and coding scheme (MCS), DL signal-to-noise ratio (SNR), reference signal received power (RSRP), uplink (UL) rate, UL BLER, and UL buffer. These names follow dataset fields; source units are not relabelled as standardized commercial KPMs.

Controlled implementation profiles: Profiles P1–P4 in Table IV are deterministic maps $\Psi_p : z_{1:T} \mapsto y_{p,1:T}$ applied to each real trace. They vary scale, unit representation, window, lag, training-scale noise, entry loss, burst loss, quantization, hidden monotone filtering, and reset semantics. P4 changes at $t_s = 0.55T$ through a persistent 1.25-standard-scale offset, a smaller nonlinear gain change, doubled noise, and two extra lag samples. No profile is attributed to a vendor.

P0 is an identity reference with 0.2% noise and is used only for software checks. P1–P3 test stationary portability, whereas P4 tests invalidation under severe unannounced drift. The global seed is 20,260,712 and each profile seed is a stable hash of $(p, \text{traceID})$.

B. xApp, Baselines, and Metrics

The fixed xApp predicts one-second quality-of-service (QoS) risk from z_t. For each traffic class, label $Y_t^{\text{risk}} = 1$ when the next-four-sample mean DL rate is below its training-only 35th percentile or BLER exceeds its training-only 75th percentile. Thresholds, feature statistics, and the class-balanced logistic model use only the first $0.6T$ samples of each canonical `exp1` trace; the four-sample horizon is removed at the boundary. The registered functional is the fitted logit

$$\ell(z) = w^\top[(z - \mu_x) \oslash \sigma_x] + b, \tag{16}$$

where (w, b) are logistic parameters, (μ_x, σ_x) are fit-prefix feature location and scale, and $\oslash$ is elementwise division. On canonical `exp2`, the instrument has area under the receiver-operating-characteristic curve (AUROC) 0.755, Brier score 0.205, and prevalence 44.9%. These quantities characterize the fixed xApp; KPM-Bridge does not alter its intrinsic accuracy.

With intervention cost $c_a = .2$ and missed-risk cost $c_m = 1$, the probability threshold is $c_a/c_m = .2$. Equivalently, (9) uses $\theta = \log(.2/.8) \simeq -1.386$ with (16). Clairvoyant-label regret is .8 for a miss and .2 for an unnecessary action. Agreement with the *same* xApp on canonical z_t is the primary portability measure, which separates telemetry translation error from source-model error.

Fitting and calibration: For each `exp1` trace, the first 60% is the fitting prefix and 10% of that prefix becomes evenly spaced paired anchors. The remaining 40% is excluded from fitting; every fourth sample enters $\mathcal{D}_v^C$. The full `exp2` repetition is untouched test data. Gradient-boosting hyperparameters are fixed across profiles.

The seven portable baselines are direct input with median fill, contract conversion, featurewise z-score alignment, CORAL [23], Gaussian OT [25], [26], supervised current-sample anchor ridge, and temporal ridge. Two bounds are also reported: deployment-specific logistic retraining on the 60% prefixes and oracle canonical input. Retraining is not portable and uses roughly ten times more task labels than the bridge uses mapping anchors.

Distribution baselines receive the same anchor-budget samples but not their eventwise pairing. With raw moments (μ_y, C_y) and canonical moments (μ_z, C_z), z-score alignment

TABLE V
INFORMATION AND SUPERVISION MADE AVAILABLE TO EACH EVALUATED METHOD FAMILY.

Method family	Contracts	Paired anchors	Task labels
Direct / distribution	no	no	no
Contract only	yes	no	no
Anchor/temporal ridge	yes	10%	no
KPM-Bridge	yes	10%	no
Deployment retrain	no	no	60% fit prefix
Oracle canonical	canonical	n/a	source model

TABLE VI
FIXED DATA SPLIT, CALIBRATION, MODEL, AND EXECUTION CONTROLS USED FOR REPRODUCIBILITY.

Item	Fixed value
Upstream revision	`bd86629d07d5`
Global seed	20,260,712
Mapper fit region	first 60% of `exp1`
Paired anchors	10% of fit region
Calibration	last 40%, stride four
Conformal level	$\alpha = 0.05$ unless swept
Residual model	70 trees/KPM, depth five
Drift anchors	every 10 samples; $W = 5$
Bootstrap	1,000 whole-trace resamples
Threads	one OpenMP/BLAS thread

is featurewise, while CORAL and Gaussian OT use

$$\begin{aligned} T_{\mathrm{CORAL}}(y) &= (y-\mu_y)C_y^{-1/2}C_z^{1/2}+\mu_z, \\ T_{\mathrm{OT}}(y) &= (y-\mu_y)C_y^{-1/2}(C_y^{1/2}C_zC_y^{1/2})^{1/2}C_y^{-1/2}+\mu_z. \end{aligned} \tag{17}$$

Both transforms use covariance regularization $10^{-5}I_d$. Anchor ridge is multivariate and supervised but receives only y_t; temporal ridge receives the same x_t as KPM-Bridge. The comparison therefore separates nonlinear residual capacity from temporal context and semantic preprocessing without withholding available history.

Table V prevents two misleading comparisons. Distribution alignment is cheaper because it is unsupervised, but it has no paired correctness signal. Deployment retraining solves a different problem: it consumes local task labels and replaces the xApp parameters (w, b). Its performance is therefore not evidence that the original model is portable.

Pooled canonical error is $\mathrm{NRMSE} = \sqrt{\mathbb{E}\|D^{-1}(\hat{z}-z)\|_2^2/d}$. Additional metrics include normalized absolute error, fixed-xApp agreement, label regret, joint coverage, radius, acceptance $A_{\mathrm{acc}} = \Pr\{G_t = 1\}$, conditional disagreement $E_{\mathrm{sel}} = \Pr\{\hat{a}_t \neq a_t^\star \mid G_t = 1\}$, and joint exposure $J_{\mathrm{exp}} = A_{\mathrm{acc}}E_{\mathrm{sel}}$, plus missed-risk rate, drift detection/delay, trace false alarm, fit time, amortized inference, and serialized bytes. Every 95% interval resamples whole test traces 1,000 times, so no row-level independence is assumed. Runtime uses one Open Multi-Processing (OpenMP)/Basic Linear Algebra Subprograms (BLAS) thread.

Every generated result carries a status distinguishing the full benchmark from the smoke test. Tables and numerical macros are produced from comma-separated outputs, and a fail-closed audit checks the remaining prose values against the same files. The manifest stores SHA-256 and byte count for every upstream input. Table VI fixes the split, seed, calibration level, model capacity, and execution environment.

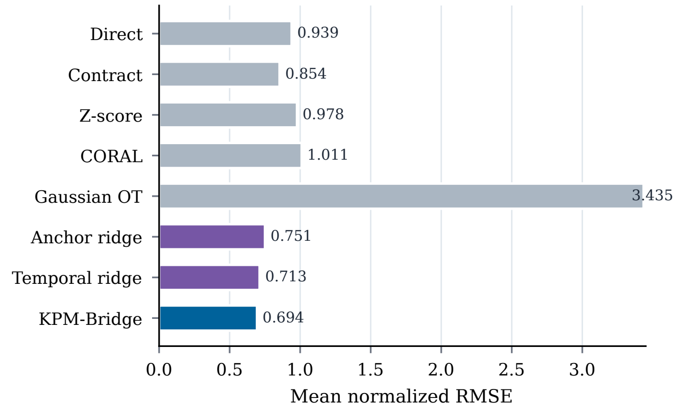

Fig. 4. Stationary normalized reconstruction error across profiles P1–P3 for all portable methods. Lower values indicate more accurate canonical KPM recovery.

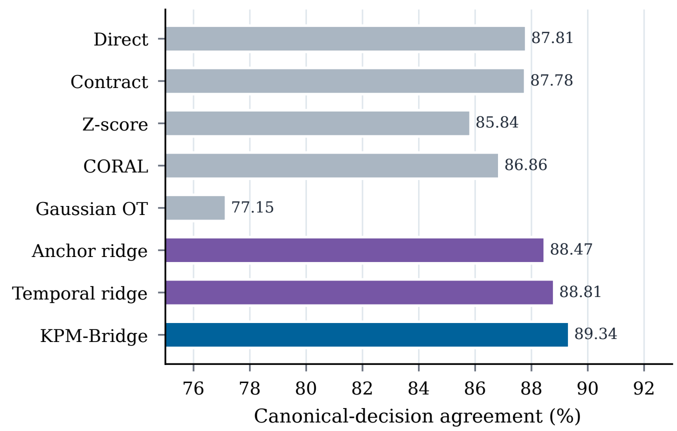

Fig. 5. Fixed-xApp decision agreement across stationary profiles P1–P3. Higher values indicate closer agreement with decisions made from canonical inputs.

VII. RESULTS

A. Stationary Cross-Implementation Portability

Figures 4 and 5 and Table VII summarize P1–P3. KPM-Bridge attains $\overline{\mathrm{NRMSE}}$ =0.694, an observed 2.6% reduction from temporal ridge, the strongest reconstruction baseline. Fixed-xApp agreement is 89.34%, or 0.53 percentage points above the strongest decision baseline. Mean label regret is 0.0977, compared with 0.1033 for CORAL and 0.1019 for temporal ridge. Because the whole-trace intervals overlap, these are descriptive differences rather than significance claims.

For P1–P3, the profile vector is $\boldsymbol{e} = (.675, .682, .725)$ with whole-trace bootstrap 95% intervals $[.600, .749]$, $[.607, .759]$, and $[.636, .817]$. Their width exposes trace heterogeneity that row-level resampling would suppress. Agreement ranges from 87.94% to 90.13%.

P2 explains much of the separation. Contract-only differentiation forms $r_t = (b_t - b_{t-1})/\Delta_t$; a missing neighbor makes r_t unavailable, while counter noise and quantization are amplified by the difference. Anchor ridge reduces the error, and temporal models use history plus m_t. Gaussian OT degrades because matching the marginal law of b_t to that of r_t cannot reconstruct eventwise increments. Distribution alignment cannot replace counter semantics.

TABLE VII
MEAN STATIONARY RECONSTRUCTION, DECISION, REGRET, AND RUNTIME PERFORMANCE ACROSS PROFILES P1–P3.

Method	NRMSE↓	Agree (%)↑	Regret↓	μs/sample
Direct	0.939	87.81	0.1097	0.03
Contract	0.854	87.78	0.1040	1.02
Z-score	0.978	85.84	0.1025	0.04
CORAL	1.011	86.86	0.1033	0.04
Gaussian OT	3.435	77.15	0.1249	0.04
Anchor ridge	0.751	88.47	0.1034	1.31
Temporal ridge	0.713	88.81	0.1019	2.34
KPM-Bridge	**0.694**	**89.34**	**0.0977**	**10.39**

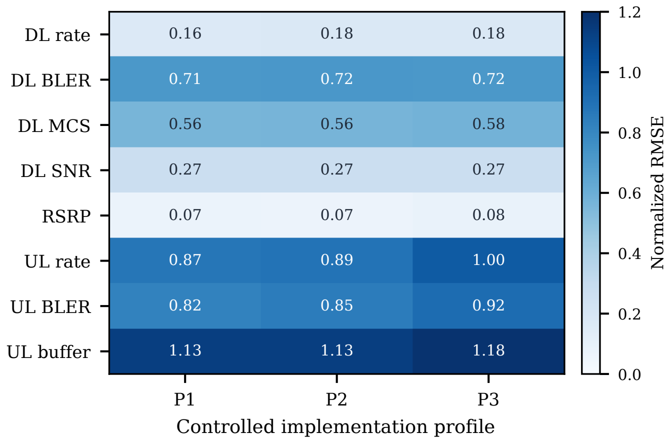

Fig. 6. Per-KPM normalized reconstruction error for KPM-Bridge across profiles P1–P3, exposing feature-level variation hidden by the pooled metric.

Direct input still obtains 87.81% agreement despite poor reconstruction because the canonical xApp acts on 88.66% of test samples. With such class imbalance, a high agreement can coexist with large $\|D^{-1}(\hat{z}-z)\|_2$. NRMSE, regret, and selective disagreement are therefore interpreted jointly.

B. Featurewise Error Structure

Figures 6 and 7 expose what the pooled norm hides. DL-rate NRMSE lies in $[.162, .176]$, and RSRP remains below .080. UL buffer is hardest, $[1.13, 1.18]$, because its distribution is zero-inflated with rare bursts; UL rate and BLER are also more variable than their DL counterparts. Relative to temporal ridge, the nonlinear residual lowers mean DL-rate error by 22.8% and RSRP error by 42.6%, but changes DL-BLER error by only 0.108%. The gain $G_k = 1 - e_k^{\text{bridge}}/e_k^{\text{ridge}}$ is thus feature dependent: nonlinear scaling and saturation benefit, whereas sparse bursts remain anchor limited.

This feature structure also explains why agreement improves less than NRMSE. In (16), the dominant coordinates are DL rate and DL BLER. KPM-Bridge materially improves the former but not the latter; RSRP and UL improvements perturb $\ell(z)$ only through smaller coefficients w_k. Another xApp, with $w' \neq w$, would require a different functional certificate and could value the same reconstruction errors differently.

C. Calibration and Selective Inference

Across P1–P3 at $\alpha = .05$, Table VIII gives empirical joint coverage 94.32%, 0.68 percentage points below $1-\alpha = .95$. Experiment-level separation removes shared repetitions, but

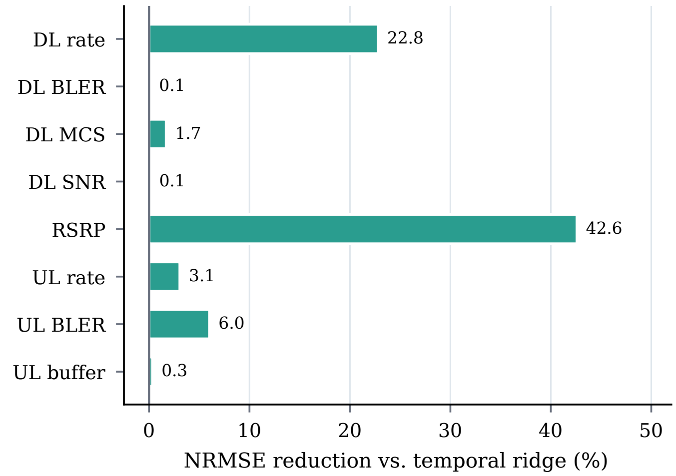

Fig. 7. Featurewise NRMSE reduction relative to temporal ridge. Positive values indicate lower reconstruction error for KPM-Bridge.

TABLE VIII
VALID-REGIME JOINT COVERAGE, WHOLE-TRACE ACCEPTANCE AND CONDITIONAL CANONICAL-ACTION DISAGREEMENT, AND DRIFT DIAGNOSTICS AT $\alpha = 0.05$.

Profile	Valid cov. (%)	Accept (%)	Cond. dis. (%)	Cluster hi.	Detect (%)	Delay (s)
P1	94.06	57.91	0.228	0.403	–	–
P2	94.45	56.78	0.202	0.361	–	–
P3	94.45	48.66	0.198	0.352	–	–
P4	94.43	35.24	6.030	8.181	88.7	8.96

temporal dependence makes exchangeability approximate. The mean generic radius is $\overline{q}_{.05} = 3.324$ training scales.

For the registered logit, I_t^ℓ is more selective than the generic d-ball. Mean acceptance is A_{acc} =54.45% and conditional canonical-action disagreement is E_{sel} =0.209%. Per-profile whole-trace upper limits are .403%, .361%, and .352%; none supports a zero-error claim. Removing $\hat{q}_\alpha^\ell$ increases E_{sel} to 9.67–11.73% while increasing A_{acc} to 78.10–92.18%.

Figures 8 and 9 sweep $(\hat{q}_\alpha, \hat{q}_\alpha^\ell)$. As $\alpha : .01 \to .20$, mean P1–P4 coverage changes $98.72\% \to 79.15\%$, acceptance $44.41\% \to 56.96\%$, and conditional disagreement $1.51\% \to 1.94\%$. The curves describe a three-way tradeoff among coverage, availability, and error; they do not determine a universal α.

D. Drift Invalidation

Before the P4 shift, empirical coverage is 94.43%. At 1.25 training scales, the residual monitor detects 88.7% of traces with 8.96 s median delay and 9.4% trace-level false alarms before the change. The denominators matter. Post shift, the full gate has $(A_{\text{acc}}, E_{\text{sel}}, J_{\text{exp}}) = (5.99\%, 76.23\%, 4.57\%)$ with $J_{\text{exp}} = A_{\text{acc}} E_{\text{sel}}$. Without drift invalidation, $(A_{\text{acc}}, E_{\text{sel}}, J_{\text{exp}}) = (98.19\%, 87.6\%, 86.0\%)$. The 94.7% exposure reduction is therefore produced mainly by $A_{\text{acc}} \downarrow$, not by making the few admitted pre-alarm actions accurate.

Figures 10 and 11 show a low-power regime: at a .5-scale shift, detection is 9.43% and post-shift coverage 90.64%. Detection rises to 71.70% at one scale and 88.68% at 1.25 scales. The false-alarm line stays at 9.43% because every sweep shares the same pre-change segment. Removing only

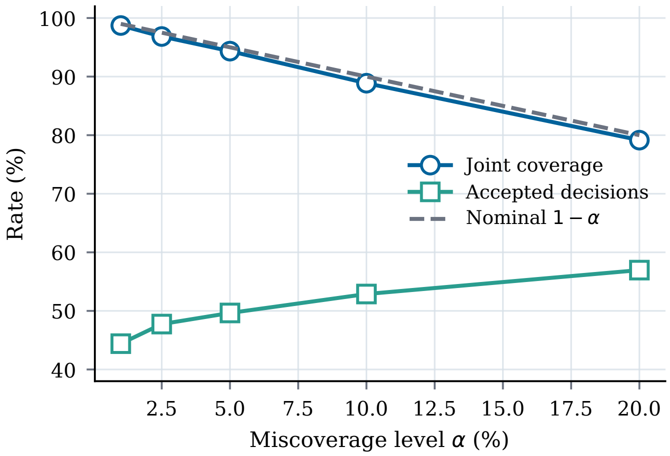

Fig. 8. Empirical joint coverage and accepted-decision rate versus conformal miscoverage level α. The dashed line is the nominal coverage target $1-\alpha$.

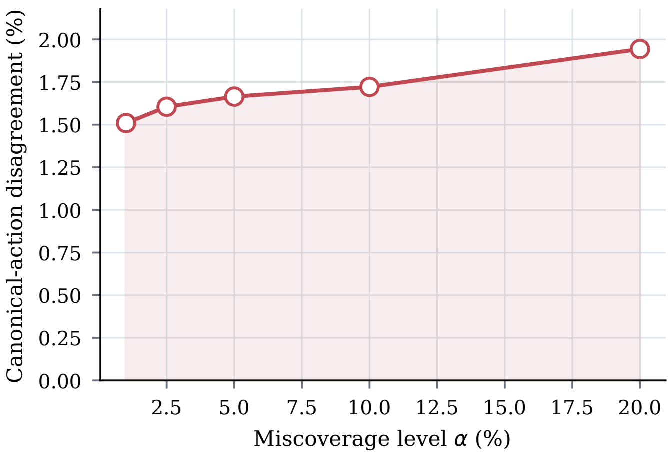

Fig. 9. Mean conditional canonical-action disagreement versus conformal miscoverage α across profiles P1–P4.

functional uncertainty changes $E : 76.23\% \rightarrow 76.14\%$ and $J : 4.57\% \rightarrow 4.73\%$. Under severe shift, I_t^ℓ cannot substitute for timely epoch invalidation.

E. Ablation and Sensitivity

Table IX gives the reconstruction sequence $.854 \rightarrow .751 \rightarrow .713 \rightarrow .694$ for contract-only, anchor ridge, temporal ridge, and the full nonlinear map. Removing typing gives .695 because the benchmark preserves column order and paired anchors. This negative result is informative: typing is justified by the invariant $T_{ik} = 0 \Rightarrow (i, k) \notin E_v$, not by manufacturing an accuracy gain on already paired columns. The executable compiler accepts declared conversions/permutations and rejects quantity and scope decoys (4/4 checks); the untyped variant cannot. The benchmark materializes 369,492 certificates, totaling $369{,}492 \times 48 = 17{,}735{,}616$ bytes.

Figures 12–14 isolate anchor fraction η_A, unseen loss p_m, and extra lag ΔL. For P3, $\eta_A : 1\% \rightarrow 5\%$ lowers NRMSE $.760 \rightarrow .724$, while 20% yields .721, showing diminishing returns in (4). A P1 mapper fitted at $p_m = .03$ is tested without refitting: agreement decreases $90.15\% \rightarrow 88.34\%$ as $p_m : 0 \rightarrow .30$ and the mask/support components of x_t deteriorate. Expected one-sample lag gives 90.13%; removing it gives

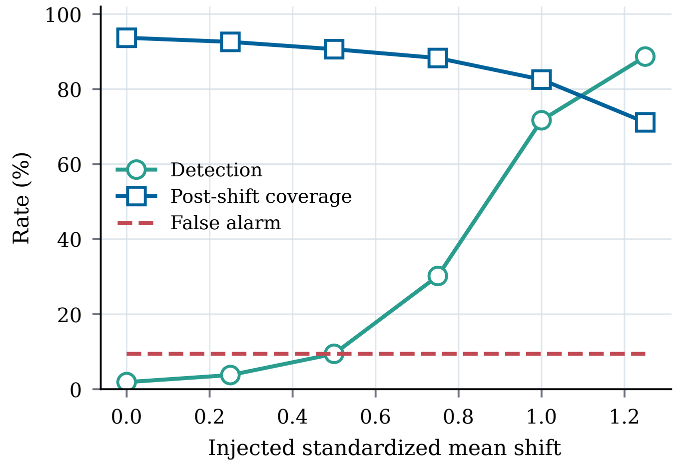

Fig. 10. Detection rate, post-shift coverage, and trace-level false-alarm rate versus the injected standardized mean shift.

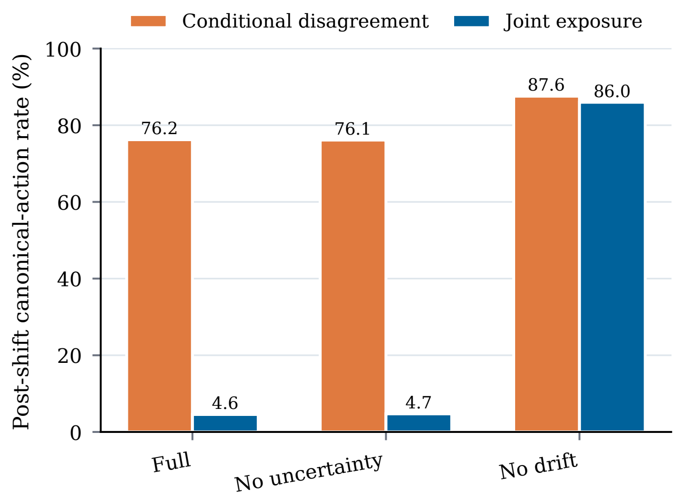

Fig. 11. Post-shift P4 conditional canonical-action disagreement and joint disagreement exposure for the full and ablated gates.

89.21%, and $\Delta L \geq 2$ lowers agreement below 87.6%. Six samples also violate the 1.5-s age budget, so $G_t = 0$ even if M_v returns a vector.

VIII. Complexity, Overhead, and Deployment

The Python reference runs on an AMD EPYC 9V74 central processing unit (CPU) with one OpenMP/BLAS thread. A profile fit takes 1.00–1.44 s; amortized inference is 10.4 μs/sample per sample and mean storage 1.45 MiB. Temporal ridge takes 2.34 μs/sample and 13.4 KiB. Figure 15 and Table X separate observed $(T_{\text{fit}}, T_{\text{inf}}, M)$ from asymptotic scaling. E2 decoding, transport, scheduling, queueing, and actuation are excluded, so these are not worst-case deadlines.

The source cadence is $T_s = 250$ ms and the QoS horizon is $4T_s = 1$ s. Although measured batch compute is small relative to T_s, deployment requires a tail budget $T_{.99}^{\text{dec}} + T_{.99}^{\text{trans}} + T_{.99}^{\text{sched}} + T_{.99}^{\text{cert}} + T_{.99}^{\text{inf}} + T_{.99}^{\text{act}} \leq B_t$.

At report rate r, N UEs, and V certified views, certificate throughput is $B_\gamma = 48rNV$ bytes/s. For $(r, N, V) = (10, 100, 1)$, $8B_\gamma$ =0.384 Mbit/s; adding an eight-float32 vector gives 0.640 Mbit/s before headers. The evaluated

TABLE IX
MEAN STATIONARY ABLATION RESULTS ACROSS P1–P3, WITH THE MECHANISM RETAINED BY EACH VARIANT.

Variant	Retained mechanism	NRMSE	Agree (%)
No residual/temporal	contract normalization only	0.854	87.78
No temporal	current report only	0.751	88.47
Linear residual	ridge instead of trees	0.713	88.81
No typed contract	statistical fit; no type rejection	0.695	89.32
Full	contracts + temporal nonlinear residual	0.694	89.34

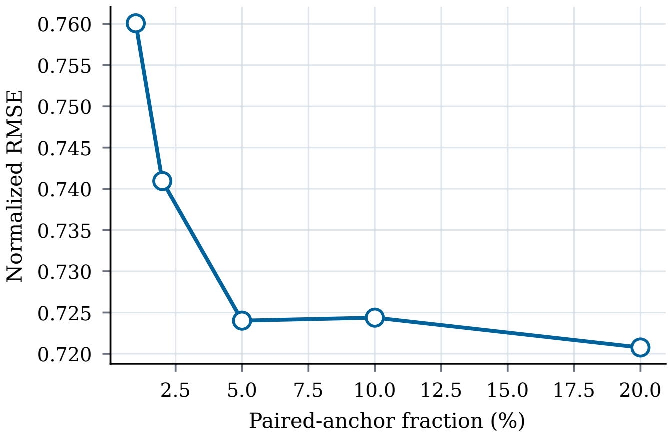

Fig. 12. P3 reconstruction error versus the paired-anchor fraction. The curve shows diminishing returns beyond the initial anchor increase.

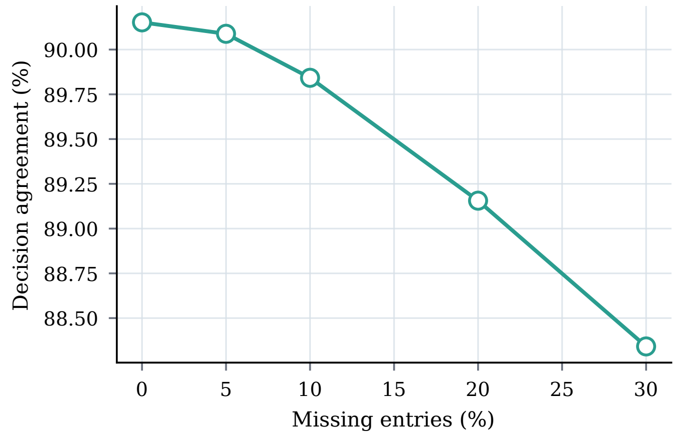

Fig. 13. Fixed-xApp decision agreement when a P1 mapper trained at 3% loss is evaluated under increasing missing-entry rates without refitting.

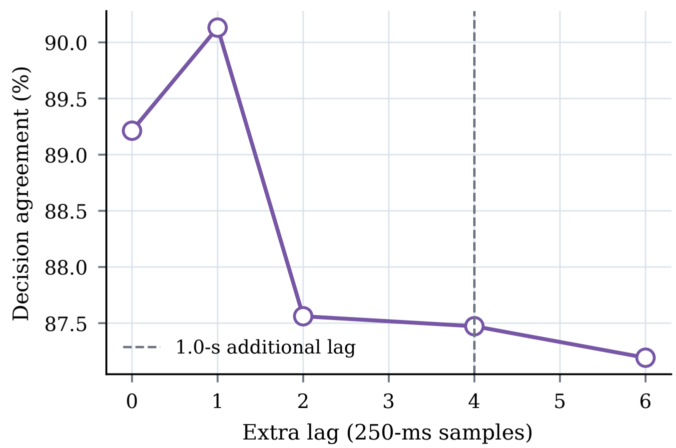

Fig. 14. Fixed-xApp decision agreement under unseen additional lag. The dashed line marks the one-second age increment reached at four 250-ms samples.

sidecar remains inside the Near-RT RIC and does not alter E2 subscriptions; external transport would add protocol and security headers.

Compilation runs only when the schema or policy version changes. Per implementation, streaming state contains nine report positions, masks, age/support, model state, and drift statistics. After compilation, work is $O(|\mathcal{V}|)$ across E2 nodes and can be sharded. Production still requires a compact tree runtime and tail-latency measurements on the target RIC.

IX. Discussion and Limitations

Certificate interpretation: The certificate binds $(C_v, \pi_v, M_v, e, \gamma_t)$ to measured operating evidence. Its finite-sample statement requires exchangeability and does not imply conditional coverage for every subgroup. Nor does $G_t = 1$ prove that the xApp is beneficial, fair, secure, or stabilizing. The object certifies telemetry fitness under a declared regime, not system trustworthiness.

Information loss and anchors: If $\ker(A) \neq \{0\}$, cell aggregation cannot identify per-UE state without side information; long windows suppress transients and consecutive losses reduce support. The correct response is a larger $\mathcal{U}_t$ or $G_t = 0$, not invented detail. A cybersecurity-driven quantum digital twin likewise maps O-RAN observations into detection registers [40]; KPM-Bridge can guard the compatibility of those inputs but cannot validate the twin's policy. The 10% anchor budget requires a trusted reference, twin, temporary dual report, or validated probe. Results remain useful at 5%, yet anchor cost and integrity are deployment constraints.

Controlled profiles versus vendors: P1–P4 are controlled transforms Ψ_p over a public experimental corpus, not measurements of commercial products. Static UEs and $T_s = 250$ ms limit generalization to mobility, handovers, multi-cell conversion, and faster loops. Vendor studies should disclose, where permitted, schemas, timestamp behavior, aggregation/reset tests, and anchor procedures.

Contract governance and versioning: Operator-governed contracts bind quantity/scope namespaces, E2SM revision, measurement identifier, labels, canonical ontology, and transform hash. A unit conversion may preserve $T_{ik} = 1$; a UE-to-cell change makes $T_{ik} = 0$; and a window or clock change invalidates epoch e. The operator, not the xApp, approves sources against model-card requirements for features, age, support, and I_t^ℓ.

Calibration under time dependence: Split-conformal coverage is finite-sample exact for exchangeable calibration units. Experiment separation and subsampling reduce but do not remove dependence. The measured 94.32% at nominal $1 - \alpha = 95\%$ motivates block-aware or weighted calibration [29].

Drift and false alarms: Without trusted anchors, H_t cannot observe drift. Under P4, detection is imperfect, 9.4% of traces alarm before the shift, and conditional post-shift disagreement remains 76.23%. The main benefit is 94.7% lower $J_{\exp} = A_{\mathrm{acc}} E_{\mathrm{sel}}$, achieved through abstention. Lowering $h_{.975}$ trades

TABLE X
ASYMPTOTIC COMPUTATION, PERSISTENT STATE, AND MEASURED REFERENCE COST BY KPM-BRIDGE LIFECYCLE STAGE.

Stage	Trigger	Time complexity	Persistent memory	Reference observation
Contract assignment	schema/policy change	$O((p_v + d)^3)$	$O(p_v d)$ costs	negligible for eight targets
Counter/event-time update	every report	$O(hd)$	$O(hd)$ history	included in mapper timing
Gradient-boosted mapping	every report	$O(dN_{\mathrm{tr}}L_{\mathrm{tr}})$	$O(dN_{\mathrm{tr}}N_{\mathrm{leaf}})$	8.2–14.1 μs/sample batched
Generic/functional gate	every report	$O(d)$	$O(1)$ quantiles	fixed 48-byte certificate
Mean-shift monitor	trusted anchor	$O(d^2 + Wd)$	$O(d^2 + Wd)$	once per 2.5 s in benchmark
Refit/recalibration	invalid epoch	offline/model dependent	calibration buffer	1.00–1.44 s profile fit

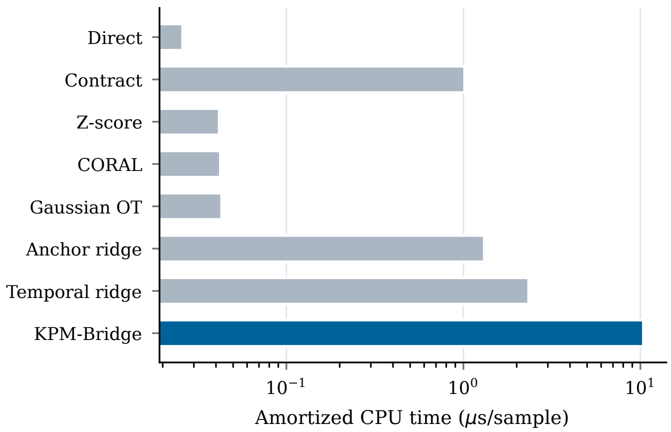

Fig. 15. Amortized single-thread processor inference time per sample for KPM-Bridge and all portable baselines; the horizontal axis is logarithmic.

delay for availability, and the per-stream-maximum threshold is empirical rather than an anytime-valid false-alarm theorem. Alarms need cause codes and operator diagnostics. Corrupting both reports and anchors can evade residual monitoring, while induced missingness can force denial of service. Authenticated anchors, execution controls, rate limits, redundant evidence, and emergency policy remain platform responsibilities.

External validity: External validation should compose xDevSM with KPM-Bridge on OAI, srsRAN, and vendor E2 traces while varying (w, τ, ρ), mobility, and schedulers. Conditional or block-aware calibration, monotone maps, and cost-aware selection of $\mathcal{D}_v^A$ remain open.

X. REPRODUCIBILITY

Source, pinned manifests, tests, and generated artifacts are available in the KPM-Bridge repository. One command regenerates the evidence; upstream GPL-3.0 traces are not redistributed.

XI. CONCLUSION

KPM-Bridge separates successful E2/API decoding from the stronger claim that $y_{v,t}$ is a valid input for a fixed xApp. Typed matching prevents quantity and scope substitution, the causal map repairs declared counter and temporal differences, $(\mathcal{U}_t, I_t^\ell)$ expose uncertainty, and H_t invalidates a stale epoch. On independent ColO-RAN repetitions, the observed stationary NRMSE is lower and canonical-decision agreement higher than for the evaluated portable baselines, with overlapping whole-trace intervals reported explicitly. Under severe drift, fail-closed gating reduces joint disagreement exposure J_{exp} by 94.7%, but accepts only 5.99% of post-shift samples; large conditional disagreement, delay, and false alarms remain. A portable xApp should therefore receive $(\hat{z}_t, \gamma_t)$, not an unqualified KPM vector.

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